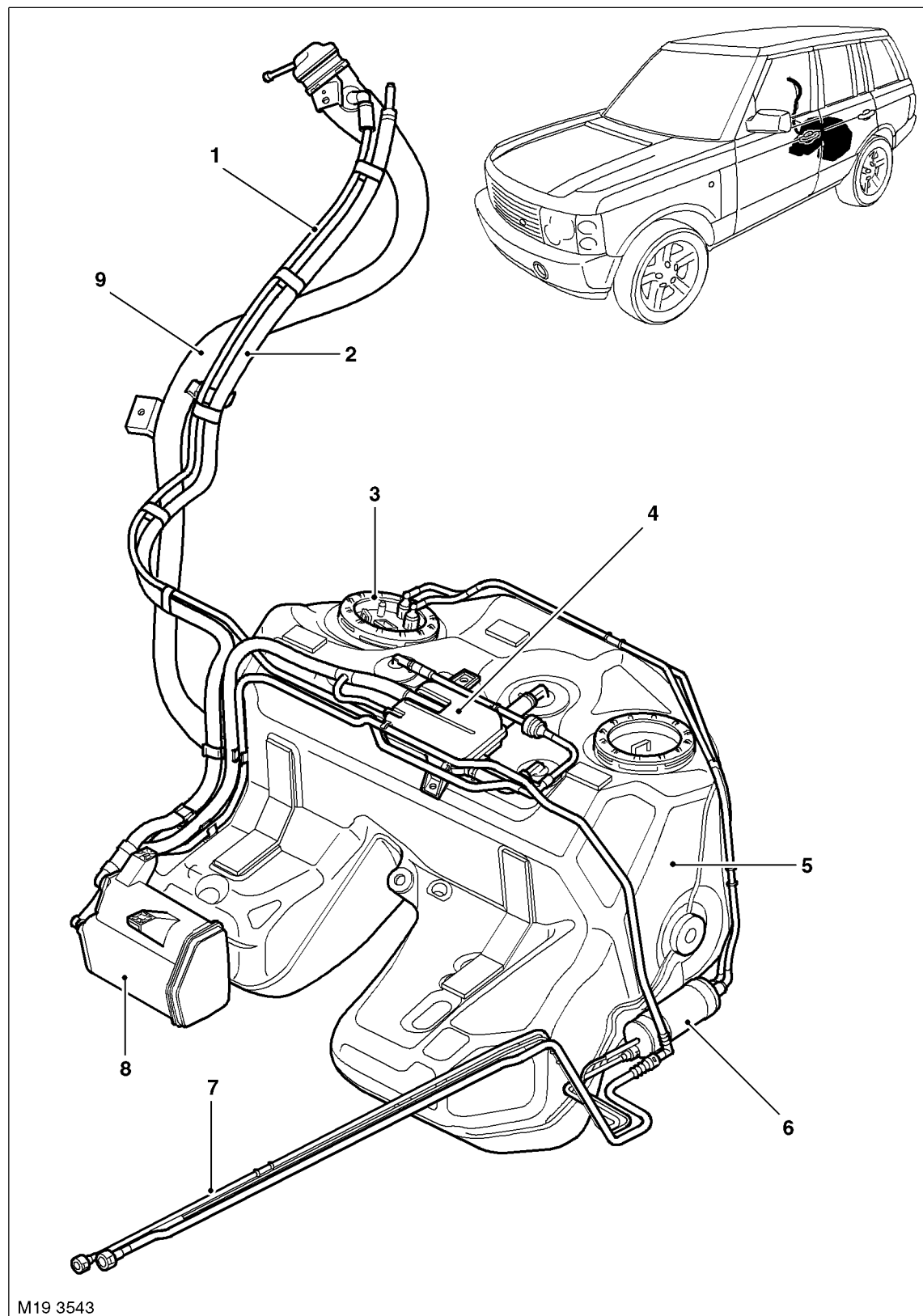


FUEL DELIVERY SYSTEM – V8

Fuel Delivery System Component Location (NAS)





- 1 Fuelling vapour recovery tube
- 2 Vent tube
- 3 Fuel pump
- 4 Vapour separator
- 5 Fuel tank
- 6 Fuel filter/pressure regulator
- 7 Fuel feed pipe
- 8 Charcoal canister
- 9 Fuel filler tube

FUEL DELIVERY SYSTEM – V8

V8 Fuel Delivery System

Fuel Delivery System Components

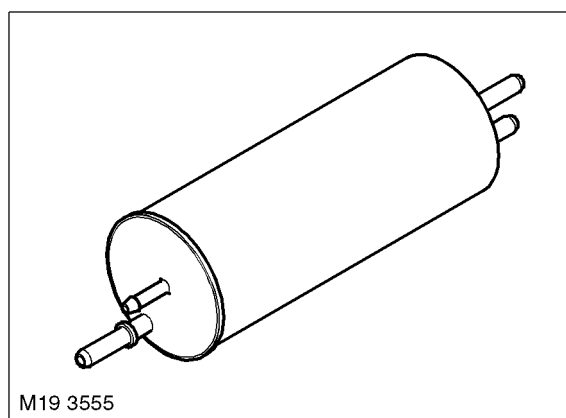
The Land Rover V8 fuel delivery system comprises the following main components:

- Saddle type fuel tank
- Fuel injectors
- Non return fuel rail
- Fuel pump
- On board refuelling vapour recovery system (NAS only)
- Tank leakage detection system (NAS only).

Fuel Tank

The fuel tank is constructed from moulded plastic and is mounted under the rear seat just forward of the rear axle. The fuel tank has a 100 Litre capacity and activates a low fuel warning lamp in the instrument pack when there is 12 litres in the tank. The tank is of the saddle tank design with a siphon jet pump in the left side to transfer fuel to the right side where the fuel pump is mounted. The fuel is delivered along pipes that run along the underside of the vehicle to a fuel rail mounted on the engine.

Fuel Filter



The fuel filter/pressure regulator is mounted on the lower right hand side of the tank. The charcoal canister is mounted at the front of the tank on the right hand side. The components are covered by a plastic under tray to protect them when driving off-road. The undertray has an inspection cover to enable access to the charcoal canister and the fuel filter.

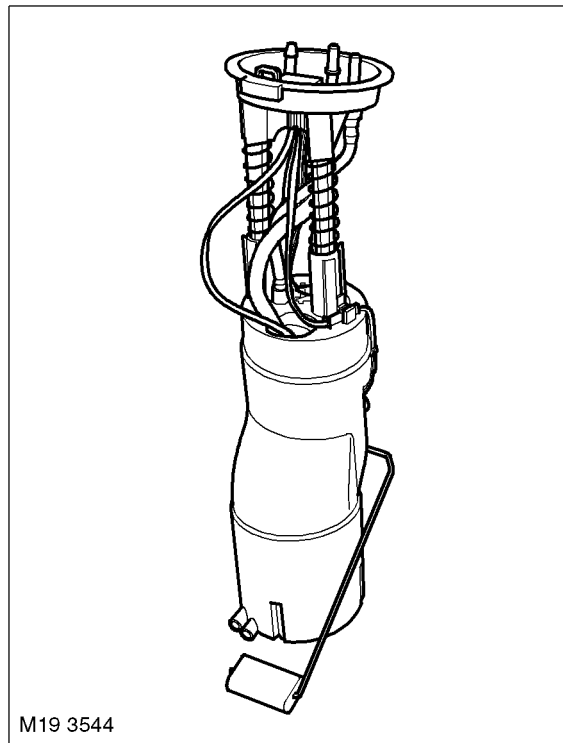
Fuel Pump

The fuel pump comprises a pump, sender unit, pot and venturi. The unit has an integral filter inside the pot. The pump unit operates at 3.5 bar for 170–205 l/hr (the V8 has a maximum operating requirement of 90 l/hr).

The pump is controlled by the ECM via the fuel pump relay.



Fuel Pump



Should the fuel pump electrical connection need to be disconnected, it is imperative that the ignition should be switched off. If the ignition is on in any position the fuel gauge will memorise its last needle position prior to power down. Once power is restored the gauge will display the last remembered position regardless of the actual level of fuel in the tank. This will result in incorrect fuel gauge readings if the fuel tank has been drained and not filled with exactly the same quantity of fuel that was removed.

Injectors

An electromagnetic, top feed fuel injector is installed in each cylinder inlet tract of the inlet manifolds. A common fuel rail supplies the injectors with fuel from a returnless fuel delivery system. The fuel in the fuel rail is maintained at a pressure of 3.5 bar by a pressure regulator incorporated in the fuel filter unit on the side of the fuel tank. A Schraeder valve is installed in the fuel rail, to the rear of injector No.7, to enable the fuel pressure to be checked.

Injector

